

Notes: Sacerdote (QJE, 2001)

Peer Effects with Random Assignment: Results for Dartmouth Roommates

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1 Big picture

- **Question.** Do randomly assigned college roommates affect one another's academic and social outcomes?
- **Core empirical advantage.** Dartmouth College randomly assigned freshmen to dorms and roommates for the graduating classes of 1997 and 1998. This removes the usual selection problem in peer-effect estimation.
- **Main outcomes.** The paper studies GPA, late graduation, major choice, fraternity/sorority/coed-house membership, fraternity-house choice, and varsity athletics.
- **Main academic finding.** Freshman roommate GPA is positively correlated with own freshman GPA. A one-standard-deviation increase in roommate GPA is associated with roughly a 0.05 increase in own GPA. When using randomized roommate background rather than roommate outcomes, having a top-quartile roommate modestly raises own freshman GPA.
- **Main social finding.** Randomly assigned roommates and dormmates affect fraternity/sorority/coed-house participation. The roommate-level effect is about 6–8 percentage points, and the dorm-level effect is larger.
- **What peer effects do not affect.** Roommates do not appear to affect choice of major or varsity athletic participation. The paper therefore shows that peer effects are outcome-specific.
- **Level of peer group.** Academic peer effects operate mostly at the room level; fraternity/social membership effects operate both at the room and dorm level.
- **Reflection problem.** Regressions of own outcomes on roommate outcomes are useful tests of peer correlation but cannot by themselves identify causal endogenous effects because roommates influence each other simultaneously. Regressions of own outcomes on randomized roommate background are cleaner causal reduced-form estimates.
- **Economic takeaway.** Even among highly selected college students, randomly assigned peers influence grades and social-group participation. But peer effects are not universal: they depend on the type of decision and on the relevant social reference group.

2 Data and measurement

2.1 Institutional setting: Dartmouth random roommate assignment

- Dartmouth College is a highly selective liberal arts institution in New Hampshire.
- Beginning with a policy change in 1993, incoming freshmen were randomly assigned to dorms and roommates.
- Freshmen were placed with other freshmen, but there were no exclusively freshman dorms.
- The Office of Residential Life assigned rooms using a blocking procedure based on housing-form answers.
- The housing-form variables include whether the student smokes, whether the student is neat versus messy, whether the student stays up late, and whether the student listens to music while studying.
- Randomization is therefore conditional on the blocks formed from these housing responses.

Interpretation: The design resembles a randomized experiment with stratification. The blocking variables are not a nuisance; they define the assignment mechanism. Therefore the regressions control for housing-form blocks or dummies so that the comparison is made within assignment strata.

2.2 Sample construction

- The paper uses Dartmouth graduating classes of 1997 and 1998 because these cohorts were subject to randomized roommate assignment.
- The starting sample contains 2,181 students.
- The author drops students in singles, students in the substance-free dorm, students without housing forms, and students who were able to request a specific roommate.
- The final baseline sample contains 1,589 students.
- Room composition in the final sample: 53 percent in doubles, 44 percent in triples, and the remainder in quads.
- Only about 3 percent of students switch roommates during freshman year. The analysis uses the original randomized freshman fall assignment.
- In rooms with more than one roommate, roommate variables are averaged.

Interpretation: The restrictions keep the sample close to the experimental assignment protocol. Dropping students who requested specific roommates is crucial because roommate choice would reintroduce selection.

2.3 Pretreatment characteristics

- Academic variables include SAT math, SAT verbal, high-school class rank, private-high-school indicator, and home state.
- The main ability control is Dartmouth’s admissions *academic index*.
- The academic index is a weighted average of SAT I, SAT II, and a converted rank score based on high-school rank and high-school size.
- Additional pretreatment data come from the Survey of Incoming Freshmen, including prior beer drinking and expected probability of graduating with honors.
- Housing-form variables are also pretreatment variables and are used to construct assignment blocks.

Interpretation: These variables serve two roles: they test whether random assignment balanced roommates’ backgrounds, and they measure contextual peer effects from roommate background.

2.4 Outcome variables

- Academic outcomes: freshman GPA, sophomore GPA, junior GPA, senior GPA, and late graduation.
- Social outcomes: membership in a fraternity, sorority, or coed Greek house; whether roommates choose the same house; varsity athletics.
- Choice outcomes: economics major, social science major, natural science major, and humanities major.
- GPA is observed term by term and aggregated by year.
- Social organization membership is ultimately decided after freshman year, which is useful because the initial roommate relationship precedes the decision.

Interpretation: The paper intentionally studies multiple outcomes. This lets the author show not only where peer effects exist, but also where they are absent.

2.5 Balance tests and random-assignment validation

- The paper regresses own pretreatment characteristics on roommate pretreatment characteristics.
- The balance tests include SAT math, SAT verbal, academic index, high-school class rank, and missing-rank indicators.
- All regressions include the housing-form block controls.
- The joint F-test for roommate background variables has p-value 0.78 in the main balance table.
- Without blocking dummies, the p-value remains non-significant at 0.48.

Interpretation: The balance test supports the identifying assumption that, conditional on the assignment blocks, roommate characteristics are as good as randomly assigned.

3 Models and empirical specifications

3.1 The peer-effects problem

The general peer-effects problem is that own outcomes may be correlated with peer outcomes for three reasons:

- **Selection:** similar students choose each other or are grouped together nonrandomly.
- **Contextual effects:** peer background characteristics affect own outcomes.
- **Endogenous effects/reflection:** peer current outcomes or behaviors affect own current outcomes and are simultaneously determined.

Interpretation: The Dartmouth setting removes selection into roommate pairs, but it does not by itself solve the reflection problem for regressions of own GPA on roommate GPA.

3.2 Structural academic model

For two roommates i and j , the paper writes a simplified simultaneous model:

$$GPA_i = \delta + \alpha(ACA_i + \mu_i) + \beta(ACA_j + \mu_j) + \gamma GPA_j + \varepsilon_i, \quad (1)$$

$$GPA_j = \delta + \alpha(ACA_j + \mu_j) + \beta(ACA_i + \mu_i) + \gamma GPA_i + \varepsilon_j. \quad (2)$$

- ACA_i is the observed academic index for student i .
- μ_i is measurement error or unobserved ability.
- β captures contextual effects from roommate ability.
- γ captures endogenous effects from roommate GPA.

Interpretation: The simultaneous form shows why regressing own GPA on roommate GPA creates reflection. Roommate GPA is both a cause and an effect inside the pair.

3.3 Reduced-form estimating equation

Substituting one roommate equation into the other gives a reduced-form model:

$$GPA_i = \pi_0 + \pi_1 ACA_i + \pi_2 ACA_j + \eta_i. \quad (3)$$

The paper estimates this model with OLS and interprets π_2 as a reduced-form effect of randomized roommate background on own GPA.

Interpretation: Because roommate academic background is randomly assigned, π_2 is not driven by selection. It captures the total effect of roommate background, combining contextual channels and any induced endogenous channels.

3.4 Outcome-on-outcome specifications

The paper also estimates regressions of the form:

$$Y_i = \alpha + \rho Y_j + X_i' \delta + X_j' \phi + B_i' \lambda + u_i, \quad (4)$$

where Y_i is own outcome, Y_j is average roommate outcome, X_i and X_j are own and roommate background controls, and B_i are housing block controls.

Interpretation: The coefficient ρ is not a clean structural peer effect when Y_i and Y_j are simultaneously determined. But under random assignment, the null of no peer effects predicts no relationship between own and roommate outcomes, so these regressions are informative tests of peer correlation.

3.5 Dorm, floor, and aggregation specifications

For social outcomes the paper estimates models such as:

$$Pr(Social_i = 1) = F(\alpha + \beta_1 Social_{roommate(i)} + \beta_2 Social_{dorm, -room(i)} + X_i' \gamma + B_i' \lambda). \quad (5)$$

- $Social_{roommate(i)}$ is the roommate's social participation.
- $Social_{dorm, -room(i)}$ is dorm average participation excluding own room.
- The paper reports probit marginal effects for social outcomes.

Interpretation: This specification asks what the relevant peer group is. Academic outcomes appear to depend on the room; social membership depends on the dorm as well.

3.6 Nonlinear and interaction specifications

To test heterogeneity, the paper divides academic index into bottom 25 percent, middle 50 percent, and top 25 percent groups. It estimates:

$$GPA_i = \alpha + \sum_{a \in A} \sum_{b \in A} \theta_{ab} \mathbf{1}\{Own_i = a, Roommate_i = b\} + B_i' \lambda + e_i, \quad (6)$$

with middle-own/middle-roommate as the omitted category.

Interpretation: This specification allows peer effects to differ by the match between own ability and roommate ability. It addresses whether sorting high- and low-ability students differently could raise total GPA.

4 Identification strategy

4.1 Step 1: use randomized roommate assignment to remove selection

- Standard peer-effect studies struggle because students choose neighborhoods, schools, peer groups, or roommates.
- Dartmouth’s randomized roommate assignment makes roommate background independent of own background conditional on the housing blocks.
- Therefore, roommate pretreatment characteristics can be used as exogenous peer shifters.

Interpretation: This is the main identification advantage of the paper. It converts peer estimation from an observational selection problem into a randomized-assignment problem.

4.2 Step 2: validate randomization with balance tests

- Table II checks whether own SAT scores, academic index, and high-school rank predict roommate characteristics.
- The joint test does not reject balance.
- Earlier nonrandomized classes show strong correlation between own and roommate backgrounds, which highlights why the randomized classes are special.

Interpretation: The balance tests do not prove perfect randomization, but they strongly support the design. They also show why the same empirical exercise would be biased in nonrandomly assigned cohorts.

4.3 Step 3: distinguish reduced-form peer effects from reflection

- Regressions of own GPA on roommate GPA are affected by reflection.
- Regressions of own GPA on roommate academic index or roommate expected honors are reduced-form causal estimates because roommate background is predetermined and randomly assigned.
- The paper reports both because outcome-on-outcome correlations are useful but must be interpreted cautiously.

Interpretation: The paper is careful: it does not overclaim that the GPA-on-GPA coefficient is a structural endogenous peer effect. The cleaner causal evidence comes from randomized roommate background.

4.4 Step 4: rule out common location shocks

- Dorm fixed effects address common dorm-level shocks such as dorm quality, noise, or culture.
- A random nonroommate from the same floor is used as an additional placebo-style check.
- The GPA relationship appears at the room level, not at the dorm or random-floor-peer level.

Interpretation: If the GPA result were driven by dorm shocks, dorm fixed effects or floor-placebo relationships would absorb it. The evidence supports roommate-level peer effects for grades.

4.5 Step 5: identify the relevant peer group for each outcome

- GPA effects appear mostly at the individual room level.
- Fraternity/sorority/coed-house membership has room-level and dorm-level effects.
- Major choice shows no meaningful roommate-level correlation.

Interpretation: Peer effects are not one-size-fits-all. Different outcomes have different social reference groups and different timing.

4.6 Threats and limitations

- The sample is one highly selective college, so external validity is limited.
- Roommate assignment is random only after conditioning on housing-form blocks.
- Some students were dropped because they requested roommates or entered special housing.
- Mechanisms are not directly identified: peer effects could arise through studying, effort norms, information sharing, friendship networks, or social pressure.
- Structural separation between contextual and endogenous peer effects requires stronger assumptions than the reduced-form design.

Interpretation: The paper is strongest as evidence that peer effects exist in a randomized college-roommate setting. It is less definitive about mechanisms and about generalizing to younger students or less selective schools.

5 Tables and figures

5.1 Table I: Summary Statistics for Sample of Dartmouth Roommates Graduating Classes of 1997 and 1998; Figure I: Distribution of Freshman Year GPA

Read:

- Table I shows the final sample of 1,589 students.
- Mean freshman GPA is 3.20, and average GPA rises to 3.41 by senior year.
- The average roommate freshman GPA is 3.19, close to own GPA because roommate variables are averaged from the same randomly assigned population.
- About 49 percent join a fraternity, sorority, or coed Greek house.
- Only 3 percent graduate late; 10 percent major in economics; the broader major categories are roughly balanced across humanities, science, and social science.
- The average SAT math score is 691 and average SAT verbal is 633, reflecting a highly selected population but with meaningful variation.
- The housing variables show the assignment-block structure: 1 percent report smoking, 69 percent report being neat, 60 percent report staying up late, and 47 percent report listening to music while studying.
- Figure I shows the freshman GPA distribution, concentrated around 3.3 but with enough dispersion to study peer effects.

Interpretation: The summary statistics establish two points. First, the sample is selective, so external validity should be interpreted carefully. Second, even within Dartmouth there is substantial variation in academic background and GPA, which gives the paper statistical power to study peer effects.

Economic message: The combination of high selectivity and remaining heterogeneity makes the result important: even among already-screened elite students, peers can affect outcomes.

TABLE I
SUMMARY STATISTICS FOR SAMPLE OF DARTMOUTH ROOMMATES GRADUATING
CLASSES OF 1997 AND 1998

Variable	Obs.	Mean	Std. dev.	Min	Max
freshman year GPA	1589	3.20	0.43	0.67	4.00
sophomore year GPA	1552	3.28	0.44	0.30	4.00
junior year GPA	1529	3.35	0.45	0.60	4.00
senior year GPA	1508	3.41	0.45	0.50	4.00
roommate freshman year GPA	1589	3.19	0.39	1.15	4.00
fraternity/sorority/coed house	1589	0.49	0.50	0.00	1.00
graduate late	1589	0.03	0.18	0.00	1.00
economics major	1589	0.10	0.31	0.00	1.00
social science major	1589	0.33	0.47	0.00	1.00
science major	1589	0.29	0.45	0.00	1.00
humanities major	1589	0.35	0.48	0.00	1.00
black	1589	0.05	0.22	0.00	1.00
SAT Math	1589	691.26	67.08	420.00	800.00
SAT Verbal	1589	632.86	70.07	360.00	800.00
academic score (incoming)	1589	204.20	12.88	151.00	231.00
high school class rank (incoming)	993	9.14	12.27	1.00	75.00
high school class rank missing	1589	0.38	0.48	0.00	1.00
private high school	1589	0.11	0.32	0.00	1.00
smokes (housing form)	1589	0.01	0.12	0.00	1.00
more neat than messy (housing form)	1589	0.69	0.46	0.00	1.00
stays up late (housing form)	1589	0.60	0.49	0.00	1.00
listens to music (housing form)	1589	0.47	0.50	0.00	1.00
same roommate sophomore year	1589	0.14	0.35	0.00	1.00
HS GPA	1328	3.56	0.51	2.00	4.00
Pre-Dart: drank beer in past year	1337	0.59	0.49	0.00	1.00

Use of beer in past year is coded 0–1 as follows: 0 = not at all, occasionally or frequently = 1. Use of beer and high school GPA come from the UCLA Higher Education Research Institute's Survey of Incoming Freshman. Housing form variables come from Dartmouth's Office of Residential Life. All other data are from Dartmouth's Computing Services Group.

Sample consists of all members of the classes of 1997 and 1998 minus the following four groups: students who were assigned to classes (1990), students from whom I could not find housing forms (1990), students assigned

(a) Table I: Summary statistics

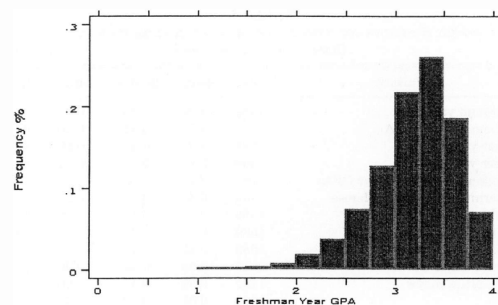


FIGURE I
Distribution of Freshman Year GPA

(b) Figure I: Distribution of freshman year GPA

5.2 Table II: Own Pretreatment Characteristics Regressed on Roommate Pretreatment Characteristics - Evidence of the Random Assignment of Roommates

Read:

- Table II is the key randomization check.
- Own SAT math, SAT verbal, academic index, and high-school rank are regressed on roommate background variables and housing-question dummies.
- Individual roommate coefficients are small and statistically insignificant.
- The joint test for all roommate background coefficients has $F(5, 1543) = 0.50$ and $p = 0.78$.
- The sample sizes are 1,589 for most columns and 993 for high-school rank, reflecting missing rank information.

Interpretation: If roommate assignment were not random, own ability would likely predict roommate ability. Table II shows no such relationship conditional on assignment blocks. This supports the central identifying assumption.

Economic message: The table is the design foundation. Without this balance evidence, later peer-effect coefficients could be confounded by sorting.

TABLE II
OWN PRETREATMENT CHARACTERISTICS REGRESSED ON
ROOMMATE PRETREATMENT CHARACTERISTICS
EVIDENCE OF THE RANDOM ASSIGNMENT OF ROOMMATES

	(1) SAT Math (self)	(2) SAT Verbal (self)	(3) HS Academic class index	(4) HS Rank	(5) HS Academic index
roommates' math SAT scores	-0.025 (0.028)				-0.005 (0.008)
roommates' verbal SAT scores		-0.009 (0.029)			-0.005 (0.007)
roommates' HS academic scores			0.010 (0.028)		0.055 (0.056)
roommates' HS class ranks				-0.032 (0.028)	0.031 (0.042)
roommates' HS class rank missing					-0.512 (0.838)
Dummies for housing questions	yes	yes	yes	yes	yes
F-test: All roommate background coeff = 0					F(5, 1543) = 0.50 P > F = .78
R ²	.09	.03	.04	.03	.04
N	1589	1589	1589	993	1589

Standard errors are in parentheses. In cases with more than one roommate, roommate variables are averaged.

Columns (1)–(5) are OLS. All regressions include 41 dummies representing nonempty blocks based upon responses to the housing questions. The lack of statistical significance on the coefficients is intended to demonstrate that the assignment process resembles a randomized experiment. In earlier nonrandomly assigned classes (such as the classes of 1995–1996), own and roommate background are highly correlated.

roommate. Regression (1) is an OLS regression of own math SAT

5.3 Table III: Peer Effects in Academic Outcomes

TABLE III
PEER EFFECTS IN ACADEMIC OUTCOMES

	(1) Fresh year GPA	(2) Fresh year GPA w/ dorm f.e.	(3) Senior year GPA	(4) Fresh year GPA	(5) Fresh year GPA	(6) Fresh year GPA	(7) Graduate late	(8) Econ major
Roommates' GPA	0.120** (0.039)	0.068** (0.029)	0.008 (0.026)					
HS academic score (self)	0.014** (0.0008)	0.015** (0.0007)	0.019** (0.0009)				-0.0001 (0.0003)	0.003** (0.0006)
HS academic score (roommates')	-0.001 (0.001)	-0.0003 (0.0009)	0.0009 (0.001)				0.0003 (0.0003)	-0.0001 (0.0006)
roommates' academic score bottom 25 percent				0.016 (0.028)	0.014 (0.025)	0.017 (0.025)		
roommates' academic score top 25 percent				0.060** (0.028)	0.047* (0.026)	0.043* (0.026)		
roommates' intention to graduate whonors (1-4)						0.082** (0.037)		
own academic score bottom 25 percent					-0.284** (0.025)	-0.282** (0.025)		

own academic score top 25 percent	0.174** (0.025)	0.175** (0.025)						
Roommate graduate late			0.008 (0.029)					
Roommate econ major							-0.018 (0.026)	
Dummies for housing questions	yes	yes	yes	yes	yes	yes	yes	yes
F test of roommate background coefficient = 0	F = 2.31 P = 0.10	F = 1.63 P = 0.20	F = 2.74 P = 0.04					
R ²	.24	.38	.19	.05	.19	.19	.06	.07
N	1589	1589	1441	1589	1589	1589	1589	1589

Standard errors are in parentheses and are corrected for clustering at the room level. In cases with more than one roommate, roommate variables are averaged. ** = p -value < .05. * = p -value < .10.
Regression (1) is OLS of own GPA on roommate GPA and controls. If own and roommate academic indices are excluded, the coefficient on roommate GPA falls to .111, and the standard error falls to 0.037.
Regression (2) adds dorm fixed effects. The coefficient on roommate GPA falls, but remains significant. Regression (3) is OLS of own senior year GPA on freshman year roommate's senior year GPA. Senior year GPA includes all grades in final year and excludes grades from earlier years.
Regressions (4)–(6) are OLS of own GPA on own and roommate background. These regressions use dummies for own and roommate academic index are in the bottom 25 percent, middle 25 percent (excluded category), or top 25 percent of their respective distributions. Regression (4) shows that "roommate top 25 percent" is significant in predicting own GPA. The level of significance on "roommate top 25 percent" falls to .10 when two dummies for own academic index are added. (This is regression (5).) Regression (6) shows that roommate intention to graduate with honors also predicts own GPA. This variable is a self-reported probability of graduating with honors and is coded as a 1, 2, 3, or 4 for the responses of no chance, very little chance, some chance, or a very good chance. Regression (6) also includes a dummy for "roommate intend to graduate with honors" missing. See text for more discussion of this variable.
Regressions (7) and (8) are probits of own "graduate late" and own "major choice = econ" on roommate graduate late and roommate major choice = econ. $1/1x$ is shown.

Read:

- Column (1) regresses own freshman GPA on roommate freshman GPA; the coefficient is 0.12 with a t -statistic of 3.1.
- A one-standard-deviation increase in roommate GPA implies about a 0.05 increase in own GPA.
- Column (2) adds dorm fixed effects; the coefficient remains statistically significant, weakening the common-dorm-shock interpretation.
- Column (3) shows that the freshman-roommate GPA effect does not persist into senior-year GPA.
- Columns (4)–(6) use randomized roommate background variables. A top-quartile roommate modestly raises own freshman GPA; the bottom-quartile roommate effect is small and insignificant.
- The own academic index dummies are strong predictors of own GPA. The top-own-index coefficient is much larger than the roommate-top-index coefficient.

- The roommate's pre-enrollment expectation of graduating with honors positively predicts own GPA in one specification.
- Graduate-late and major-choice regressions show no strong roommate effect.

Interpretation: Table III separates three empirical objects. Outcome-on-outcome GPA regressions show strong peer correlation but face reflection. Background-on-outcome regressions are cleaner causal reduced forms. The major and late-graduation columns show that roommate effects are not universal.

Economic message: The academic peer effect is meaningful but modest. It is strongest for short-run GPA and weaker or absent for longer-run decisions.

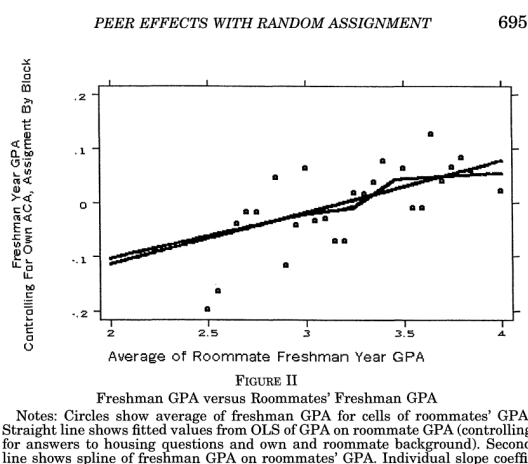
5.4 Figure II: Freshman GPA versus Roommates' Freshman GPA

Read:

- Figure II plots cell averages of own freshman GPA against average roommate freshman GPA.
- The straight fitted line is upward sloping.
- A spline fit lies close to the linear fit, and the spline segment slopes are not statistically different.
- The figure controls for housing-question blocks and own/roommate background variables.

Interpretation: The graph visualizes the Table III GPA relationship. It also suggests little evidence for strong nonlinearities in the GPA-on-roommate-GPA relationship.

Economic message: Peer effects in grades appear approximately linear over the relevant Dartmouth roommate GPA range.



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TABLE IV
OWN AND ROOMMATE MAJOR CHOICE COMPARED WITH NULL HYPOTHESIS OF NO CORRELATION IN MAJOR CHOICE
BOLD SHOWS FRACTION OF SAMPLE IN EACH CELL.
ITALICS SHOWS EXPECTED FRACTION IF OWN CHOICE AND ROOMMATE CHOICE ARE INDEPENDENT (STANDARD ERROR UNDER NULL OF INDEPENDENCE IS SHOWN IN PARENTHESES).

	Roommate Division of Major			
	humanities	sciences	social sciences	total
Own division of major				
humanities	0.13 <i>0.12</i> <i>(0.01)</i>	0.11 <i>0.11</i> <i>(0.01)</i>	0.11 <i>0.12</i> <i>(0.01)</i>	0.35
sciences	0.11 <i>0.11</i> <i>(0.01)</i>	0.10 <i>0.10</i> <i>(0.01)</i>	0.11 <i>0.11</i> <i>(0.01)</i>	0.32
social sciences	0.12 <i>0.12</i> <i>(0.01)</i>	0.11 <i>0.11</i> <i>(0.01)</i>	0.11 <i>0.11</i> <i>(0.01)</i>	0.33
total	0.35	0.32	0.33	1.00
N = 842				

Analysis done only for rooms with exactly two students.

point. I limit the sample to rooms of two. I compare the fraction of roommate pairs with the same major to the fraction that would be expected under a null of independence across roommates. For

5.5 Table IV: Own and Roommate Major Choice Compared with Null Hypothesis of No Correlation in Major Choice

Read:

- Table IV restricts attention to rooms with exactly two students, giving $N = 842$.
- It compares actual joint major-choice shares with expected shares under independence.
- Humanities-humanities pairs occur at 0.13, compared with 0.12 expected under independence.
- Science-science pairs occur at 0.10, exactly matching the 0.10 expected share.
- Social-science-social-science pairs occur at 0.11, matching the expected 0.11 share.
- The table accepts the null of independence between own and roommate broad major division.

Interpretation: Major choice does not move with random roommate assignment in this sample. This is important because major choice is a more durable labor-market-relevant decision than freshman GPA or social membership.

Economic message: Peer effects may shape effort and social behavior, but not every high-stakes educational choice is socially contagious at the roommate level.

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TABLE IV
OWN AND ROOMMATE MAJOR CHOICE COMPARED WITH NULL HYPOTHESIS OF NO
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BOLD SHOWS FRACTION OF SAMPLE IN EACH CELL.
*ITALICS SHOWS EXPECTED FRACTION IF OWN CHOICE AND ROOMMATE CHOICE ARE
INDEPENDENT (STANDARD ERROR UNDER NULL OF INDEPENDENCE IS SHOWN IN
PARENTHESES).*

Roommate Division of Major				
	humanities	sciences	social sciences	total
Own division of major				
humanities	0.13 <i>0.12</i> <i>(0.01)</i>	0.11 <i>0.11</i> <i>(0.01)</i>	0.11 <i>0.12</i> <i>(0.01)</i>	0.35
sciences	0.11 <i>0.11</i> <i>(0.01)</i>	0.10 <i>0.10</i> <i>(0.01)</i>	0.11 <i>0.11</i> <i>(0.01)</i>	0.32
social sciences	0.12 <i>0.12</i> <i>(0.01)</i>	0.11 <i>0.11</i> <i>(0.01)</i>	0.11 <i>0.11</i> <i>(0.01)</i>	0.33
total	0.35	0.32	0.33	1.00
N = 842				

Analysis done only for rooms with exactly two students.

point. I limit the sample to rooms of two. I compare the fraction of roommate pairs with the same major to the fraction that would be expected under a null of independence across roommates. For

Table IV: Own and roommate major choice compared with independence

5.6 Table V: Peer Effects in Social Outcomes

Read:

- Column (1) shows that a roommate's fraternity/sorority/coed-house membership raises own membership probability by about 7.8 percentage points.
- Column (2) includes dorm average membership excluding own room. Roommate membership remains positive at about 5.6 percentage points, and dorm membership has a larger marginal effect of about 32.1 percentage points.
- Column (3) uses beer drinking before college. Own beer use strongly predicts own Greek-house membership; roommate beer use does not, but dormmate beer use is positive and significant.
- Column (4) shows no statistically significant roommate effect for varsity athletics.

- Probit marginal effects are reported, and standard errors are clustered at the room or dorm level depending on the variable.

Interpretation: Social-group participation has a wider reference group than GPA. Roommates matter, but dormmates matter more for fraternity/sorority/coed-house participation. Athletics does not show the same peer pattern, likely because varsity team participation is heavily determined before college.

Economic message: The relevant peer group depends on the activity: social organizations are influenced by dorm networks, while GPA is mainly room-level.

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TABLE V
PEER EFFECTS IN SOCIAL OUTCOMES

	(1) Member frat/ soror	(2) Member frat/ soror	(3) Member frat/ soror	(4) Varsity athlete
roommate member of fraternity/sorority/coed	0.078** (0.038)	0.056 (0.037)		
dorm average of fraternity/sorority/coed		0.321** (0.135)		
roommate varsity athlete				0.045 (0.033)
HS academic score (self)	0.0098 (0.0010)	0.0011 (0.0011)	0.0010 (0.0011)	-0.004** (0.001)
HIS academic score (roommates)	-0.0017 (0.0011)	-0.0016 (0.0011)	-0.0016 (0.0011)	-0.0002 (0.0007)
Own use of beer in high school (0-1)			0.135** (0.038)	
Roommates' use of beer in high school (0-1)			-0.025 (0.026)	
Dormmates' use of beer in high school (0-1)			0.287** (0.146)	
Dummies for housing questions	yes	yes	yes	yes
R ²	.02	.02	.03	.05
N	1589	1589	1589	1589

Standard errors are in parentheses and are corrected for clustering at the room level. In cases with more than one roommate, roommate variables are averaged. ** = p -value < .05.

Columns (1)-(4) are Probits. $\Delta y/\Delta x$ is shown.

In regression (2), dorm average of frat membership excludes own observation, and standard errors are corrected for clustering at dorm level.

In regression (3), use of beer in past year is coded 0-1 as follows: 0 = not at all, occasionally or frequently = 1. Dorm use of beer excludes own room and standard errors are corrected for clustering at dorm level.

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TABLE VI
INTERACTION BETWEEN OWN BACKGROUND AND ROOMMATE BACKGROUND
EFFECT ON OWN FRESHMAN GPA
RELATIVE TO OWN ACADEMIC INDEX = MIDDLE, ROOMMATE'S = MIDDLE

Own academic index	Roommate academic index		
	Bottom 25%	Middle 50%	Top 25%
bottom 25 percent	-0.331** (0.056)	-0.304** (0.035)	-0.160** (0.049)
middle 50 percent	0.039 (0.034)	0	-0.019 (0.036)
top 25 percent	0.146** (0.045)	0.159** (0.037)	0.243** (0.044)

Mean freshman GPA is 3.20. Standard errors are in parentheses and are adjusted for room level clustering. ** = p -value < .05. N = 1589.

Coefficients are from the following regression of GPA on (0, 1) indicator variables: $GPA_i = \beta_0 + \beta_1 * (\text{own} = \text{bottom}, \text{roommate} = \text{bottom}) + \beta_2 * (\text{own} = \text{bottom}, \text{roommate} = \text{middle}) + \beta_3 * (\text{own} = \text{bottom}, \text{roommate} = \text{top}) + \beta_4 * (\text{own} = \text{middle}, \text{roommate} = \text{bottom}) + \beta_5 * (\text{own} = \text{middle}, \text{roommate} = \text{top}) + \beta_6 * (\text{own} = \text{top}, \text{roommate} = \text{bottom}) + \beta_7 * (\text{own} = \text{top}, \text{roommate} = \text{middle}) + \beta_8 * (\text{own} = \text{top}, \text{roommate} = \text{top}) + \gamma * X + \epsilon_i$. X is a vector of dummies for the student's choices on the housing forms.

F-test on (own = bottom, roommate = bottom) = (own = bottom, roommate = top): $F(1, 704) = 6.27$, $p = 0.013$. This shows that bottom people matched with top roommates outperform bottom people matched with bottom roommates.

F-test on (own = top, roommate = top) = (own = top, roommate = bottom): $F(1, 704) = 3.31$, $p = 0.0691$. This shows that top people matched with top roommates outperform top people matched with bottom roommates. This result is significant at the .10 level.

Redistribution experiment 1. Consider two rooms. One has two top roommates, and one has two bottom roommates. Rearrange into two "mixed" rooms, each of which contains one top and one bottom person. The two top people are estimated to each lose a benefit to GPA of .10 = .24 - .14 for a combined loss of .20. The bottom people each gain .17 for a combined gain of .34 and a net "social gain" of .14. An F-test on this gain of .14 yields a p -value of .42. I find that the "redistribution" experiment does not yield statistically significant gains.

Redistribution experiment 2: Consider two rooms. Each has one middle and one top person. Rearrange such that the top people are together and the middle people are together. The top people each gain .084, and the middle people each gain .019 (insignificantly) for a net social gain of .206. The F-test on this gain has $p = 0.066$ and is significant at the .10 level.

students. Such an experiment is considered in the notes to Table VI. I consider breaking up two mixed pairs of one top and one middle student each to form two homogeneous pairs of two top students and two middle students. The top people would

5.7 Table VI: Interaction between Own Background and Roommate Background - Effect on Own Freshman GPA

Read:

- The omitted group is own academic index = middle 50 percent and roommate academic index = middle 50 percent.
- Bottom-index students with bottom-index roommates have a coefficient of -0.331; bottom-index students with top-index roommates have a coefficient of -0.160.
- The difference between those two coefficients is statistically significant with $p = 0.013$.
- Middle-index students appear largely unaffected by roommate type.
- Top-index students with top-index roommates have a coefficient of 0.243, compared with 0.146 for top-index students with bottom-index roommates.
- The top-top versus top-bottom difference is significant at the 10 percent level.
- The redistribution exercises suggest possible social gains from rearranging roommates, but the tests are suggestive rather than conclusive.

Interpretation: Table VI shows modest evidence of nonlinear match effects. Top roommates seem helpful for bottom students, and top students may also benefit from top roommates. However, the estimates are not strong enough to support a definitive assignment policy.

Economic message: The table asks whether peer effects imply optimal sorting or mixing. The answer is provocative but cautious: certain matches may raise total GPA, but the evidence is not decisive.

6 Interpretive synthesis

- **Why random assignment matters.** The paper’s central empirical advantage is not a complex estimator but a clean assignment mechanism.
- **What is causal.** Roommate background effects are credible causal reduced forms. Roommate outcome effects are peer correlations consistent with peer effects but subject to reflection.
- **What the results say about mechanisms.** GPA effects are local to the room and likely involve study habits, effort, information, or academic norms. Fraternity effects operate through dorm-level social networks.
- **Why no effect on major matters.** If roommates strongly changed all student outcomes, major choice should move as well. The absence of a major effect shows that peer influence is selective.
- **Policy interpretation.** The evidence is relevant for roommate assignment, dorm composition, and social mixing, but the paper does not fully solve the optimal assignment problem.

7 Short summary

- The paper studies peer effects among Dartmouth roommates using randomized freshman roommate assignment.
- The final sample contains 1,589 students from the classes of 1997 and 1998.
- Balance tests show no significant relationship between own and roommate pretreatment academic characteristics conditional on housing-form blocks.
- Own freshman GPA is positively correlated with roommate freshman GPA.
- Roommate top-quartile academic background modestly raises own freshman GPA.
- GPA effects are mainly room-level, not dorm-level.
- Peer effects in fraternity/sorority/coed-house membership occur at both room and dorm levels.
- Roommates do not appear to affect major choice or varsity athletic participation.
- Nonlinear interaction evidence suggests top roommates may help bottom-index students, but redistribution conclusions are tentative.
- The main contribution is credible evidence of peer effects in a randomized college residential setting.

8 Conclusion

8.1 Core conclusion

Sacerdote shows that randomly assigned college roommates affect some student outcomes. The strongest evidence is for freshman GPA and fraternity/sorority/coed-house participation. The effects are meaningful but not universal: they appear for grades and social affiliation but not for major choice or varsity athletics.

8.2 Economic intuition

Peer effects arise because students observe, interact with, and learn from their roommates and dormmates. For GPA, the roommate is the key reference group because study routines and academic effort are shaped inside the room. For social organizations, dormmates matter because social information and participation norms spread through wider residential networks.

8.3 Identification takeaway

The paper's identification rests on randomized roommate assignment, validated by balance tests. The cleanest causal estimates use randomized roommate background. Outcome-on-outcome regressions are valuable but must be interpreted as correlations under reflection.

8.4 Final takeaway

The paper is a classic demonstration that randomized peer assignment can identify social effects. It also teaches an important empirical lesson: peer effects differ by outcome, by timing, and by the relevant social group.